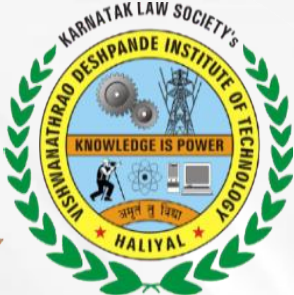




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Transforming Through Technology

**Department of
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www.klsvdit.edu.in



**Visvesvaraya Technological University, Jnana Sangama,
Belagavi – 590 018**



**KLS Vishwanathrao Deshpande Institute of Technology,
Haliyal – 581 329**

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**A Presentation entitled
Body Posture Monitor For Desk Workers**

Presented by

Srushti Kolekar

2VD23EC106

Sudeep T. Gotur

2VD23EC107

Suhas Rotti

2VD23EC108

Sujata S. Waghamode

2VD23EC110

Under the Guidance of

Dr. Nagaraj. Bhat

Department of E&CE

(Academic Year 2023-24)



PRESENTATION OUTLINE

- Motivation
- Introduction
- Literature Survey
- Objectives
- Block Diagram
- Circuit Diagram
- Working
- Outcomes
- Advantages
- Applications
- References
- Acknowledgement



MOTIVATION



- Many desk workers suffer from back pain and neck pain
- People unknowingly sit in the wrong posture for long hours
- No real-time alert system to correct posture
- Medical treatment is costly and time-consuming
- Prevention is better than a cure
- Simple technology can improve daily life
- Our goal is to use sensors to improve human health
- This project brings technology and health together



INTRODUCTION



- Body posture is the position in which we sit or stand
- Correct posture keeps the spine healthy
- Desk workers sit for long hours daily
- Wrong sitting posture causes: (Back pain, Neck pain, Shoulder strain)
- Most people do not realise when their posture is wrong
- Technology can help by detecting posture automatically
- Our project monitors posture and gives feedback using vibration alerts



LITERATURE SURVEY

SL. NO	Paper Title	Author Names	Proposed Work	Published Year
1	UWB-PostureGuard: A Privacy-Preserving RF Sensing System for Continuous Ergonomic Sitting Posture Monitoring	Haotang Li, Zhenyu Qi, Sen He, Kebin Peng, Sheng Tan, Yili Ren, Tomas Cerny, Jiyue Zhao, Zi Wang	Proposes a contactless, UWB (ultra-wideband) RF sensing system that continuously monitors sitting posture with 99.11% accuracy . It preserves user privacy (no camera/wearables) and is robust to variations like clothing, furniture, and environment.	2025
2	Sitting Posture Recognition and Feedback: A Literature Review	Christian Krauter, Katrin Angerbauer, Aimée Sousa Calepso, Alexander Achberger, Sven Mayer, Michael Sedlmair	Comprehensive literature review (N=223) across engineering, electronics, medical sciences; highlights use of pressure-sensing hardware and visual feedback; discusses factors such as environment, cost, privacy, portability, accuracy, user capability and long-term effects	2024
3	Sensor-Based Wearable Systems for Monitoring Human Motion and Posture: A Review	Xinxin Huang, Yunan Xue, Shuyun Ren, Fei Wang	Systematic review of wearable posture-monitoring systems; analyses sensor types (IMU, flex, pressure), placement strategies, system design aspects, and emphasises future needs like accuracy, data security, comfort, and durability.	2024
09/10/2026 Body Posture Monitor For Desk Workers 6				



LITERATURE SURVEY

SL.NO	PAPER TITLE	AUTHOR NAMES	PROPOSED WORK	PUBLISHED YEAR
4.	A Smart System for Sitting Posture Detection Based on Force Sensors and Mobile Application	Slavomir Matuska, Martin Paralic, Robert Hudec	Presents a smart chair with six flexible force sensors connected to an Arduino-based IoT node; detects incorrect sitting positions and sends feedback to a mobile app, with posture data stored in a private cloud and visualized via mobile UI.	2022
5.	Human Posture Detection Method Based on Wearable Devices	Xiaoou Li Yangang Li Yuanming Wu Jiahui Xu Zhitao Xiao	Proposes a multimodal wearable posture detection method using upper-limb sEMG, lower-limb inertial and plantar pressure sensors; features are extracted via wavelet transforms and classified using optimized SVM with ~90% accuracy.	2021



OBJECTIVES

- To design a posture monitoring system for desk workers.
- To measure body angle using the MPU6050 sensor.
- To process sensor data using ESP32.
- To detect wrong posture accurately.
- To alert the user using vibration feedback.
- To avoid false alerts during temporary movements.
- To implement a **40-second delay** before triggering vibration.
- To improve health and working comfort.



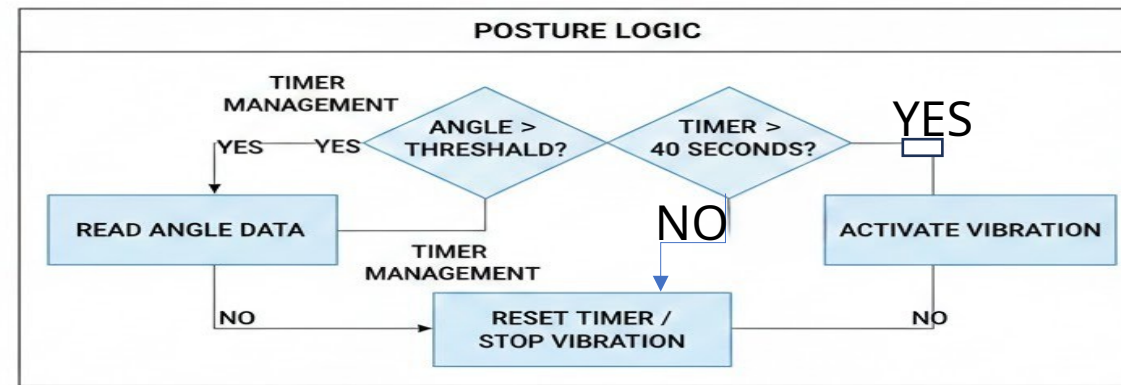
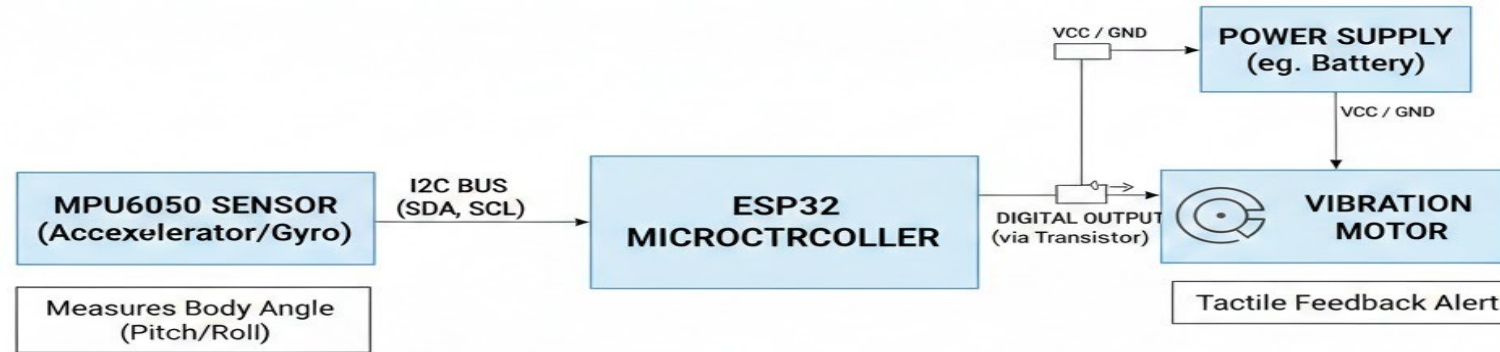
COMPONENTS

- MPU6050 IMU Sensor
- Vibration Motor
- BC547 NPN Transistor
- Resistor
- Breadboard
- Jumper Wires
- USB Cable|



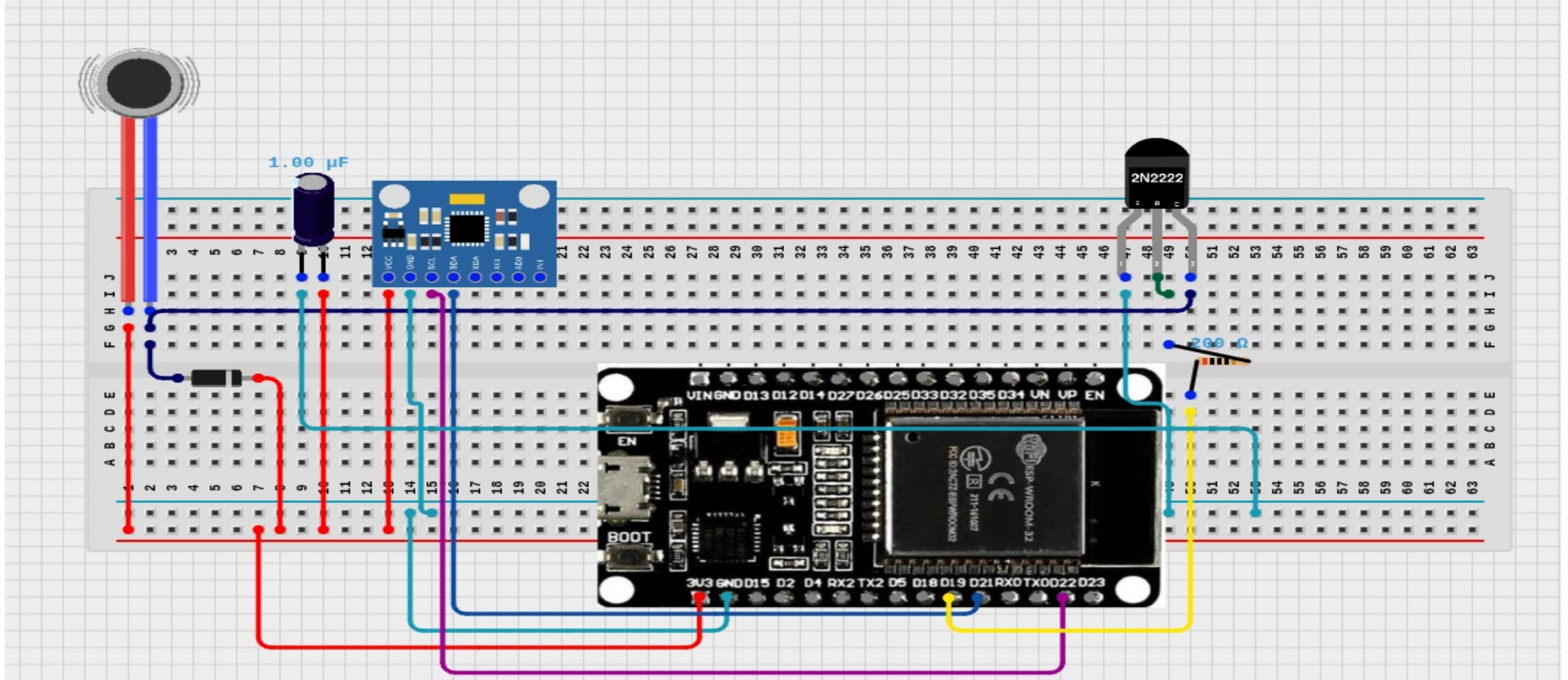
BLOCK DIAGRAM

SMART POSTURE CORRECTOR - BLOCK DIAGRAM





CIRCUIT DIAGRAM





WORKING



- The MPU6050 sensor continuously measures body angle.
- ESP32 reads angle data from MPU6050.
- The angle is compared with a preset normal posture value.
- If posture is correct → System continues monitoring.
- If posture is wrong, → Timer starts counting.
- If posture is corrected within 40 seconds, → Timer resets.
- If posture remains wrong for more than 40 seconds →
 - Vibration motor is activated.
- Vibration alert reminds user to correct posture.
- Once posture is corrected → Vibration stops.



OUTCOMES



- The system accurately detects the user's body posture.
- It continuously monitors posture in real time.
- The vibration alert effectively notifies the user when posture is incorrect.
- False alerts are reduced by using a delay timer mechanism.
- The system helps increase awareness about correct posture habits.
- Users receive a physical reminder through vibration to correct their posture.
- The system is portable and suitable for daily use.
- The project helps in reducing long-term back and neck problems.



ADVANTAGES

- The system helps users maintain correct sitting posture.
- It provides real-time feedback through vibration alerts.
- The delay timer prevents unnecessary or false alerts.
- The device is simple and easy to operate.
- The system is affordable and cost-effective.
- People of all age groups can use it.
- The device improves long-term spinal health.



APPLICATIONS

- The system can be used in corporate offices to promote healthy working posture.
- It is suitable for educational institutions to improve students' and teachers sitting habits.
- The device is useful in call centers and BPO industries.
- It can be implemented in co-working spaces.
- It is useful for people with back or neck discomfort.
- The system can be used in rehabilitation and physiotherapy centers.
- It is suitable for training programs focused on workplace safety.
- The device can be used to monitor posture during long reading or writing sessions.



REFERENCES

- Haotang Li, Zhenyu Qi, Sen He, Kebin Peng, Sheng Tan, Yili Ren, Tomas Cerny, Jiyue Zhao, Zi Wang, “UWB-PostureGuard: A Privacy-Preserving RF Sensing System for Continuous Ergonomic Sitting Posture Monitoring,” 2025.
- Christian Krauter, Katrin Angerbauer, Aimée Sousa Calepso, Alexander Achberger, Sven Mayer, Michael Sedlmair, “ Sitting posture Recognition and Feedback: A Literature Review.”:2024.
- Xiaou Li, Yangang Li, Yuanming Wu, Jiahui Xu, Zhitao Xiao,” Human Posture Detection Method Based on Wearable Devices”:2021
- Slavomir Matuska, Martin Paralič, Robert Hudec, “A Smart System for Sitting Posture Detection Based on Force Sensors and Mobile Application”, 2022



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THANK YOU